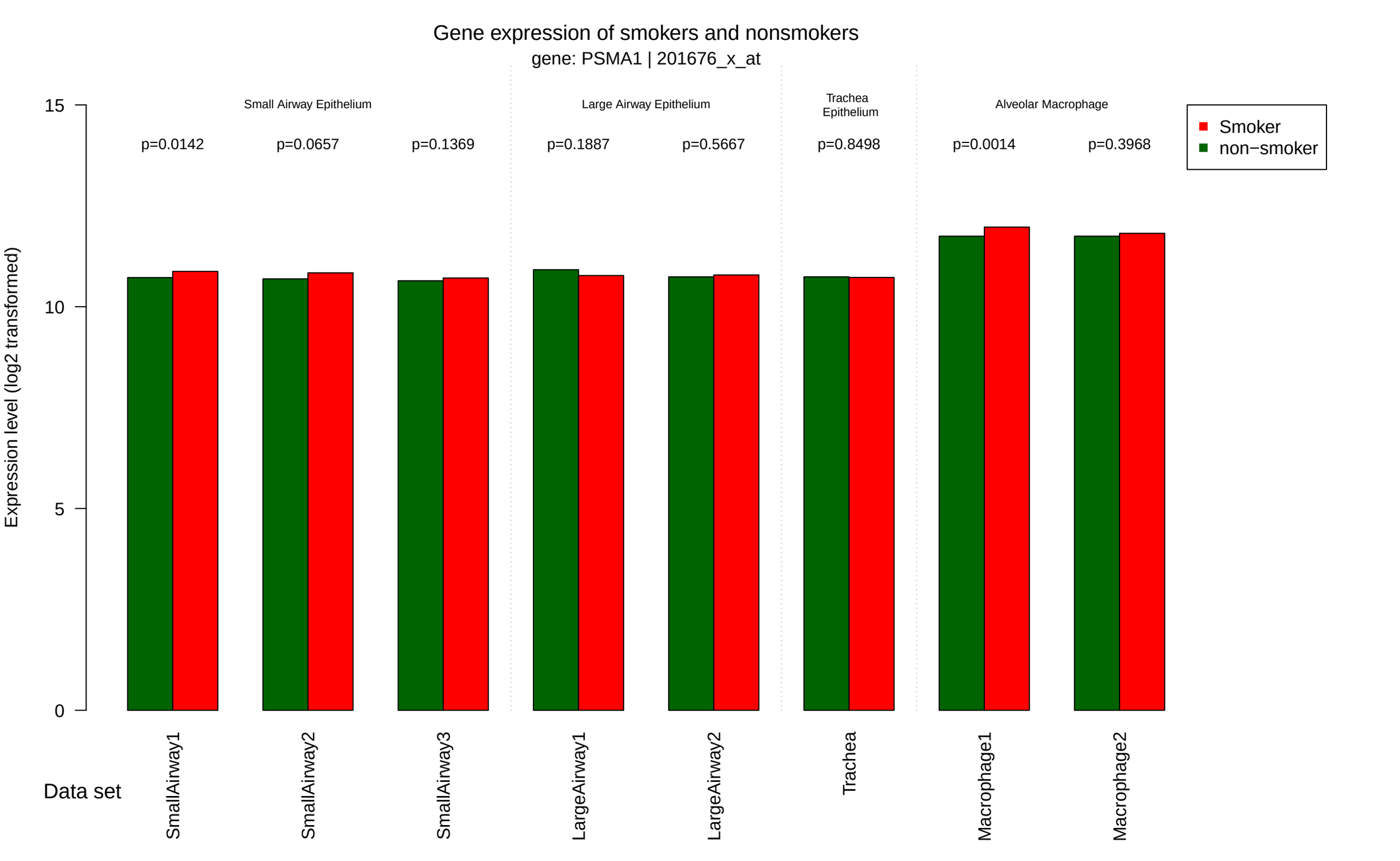
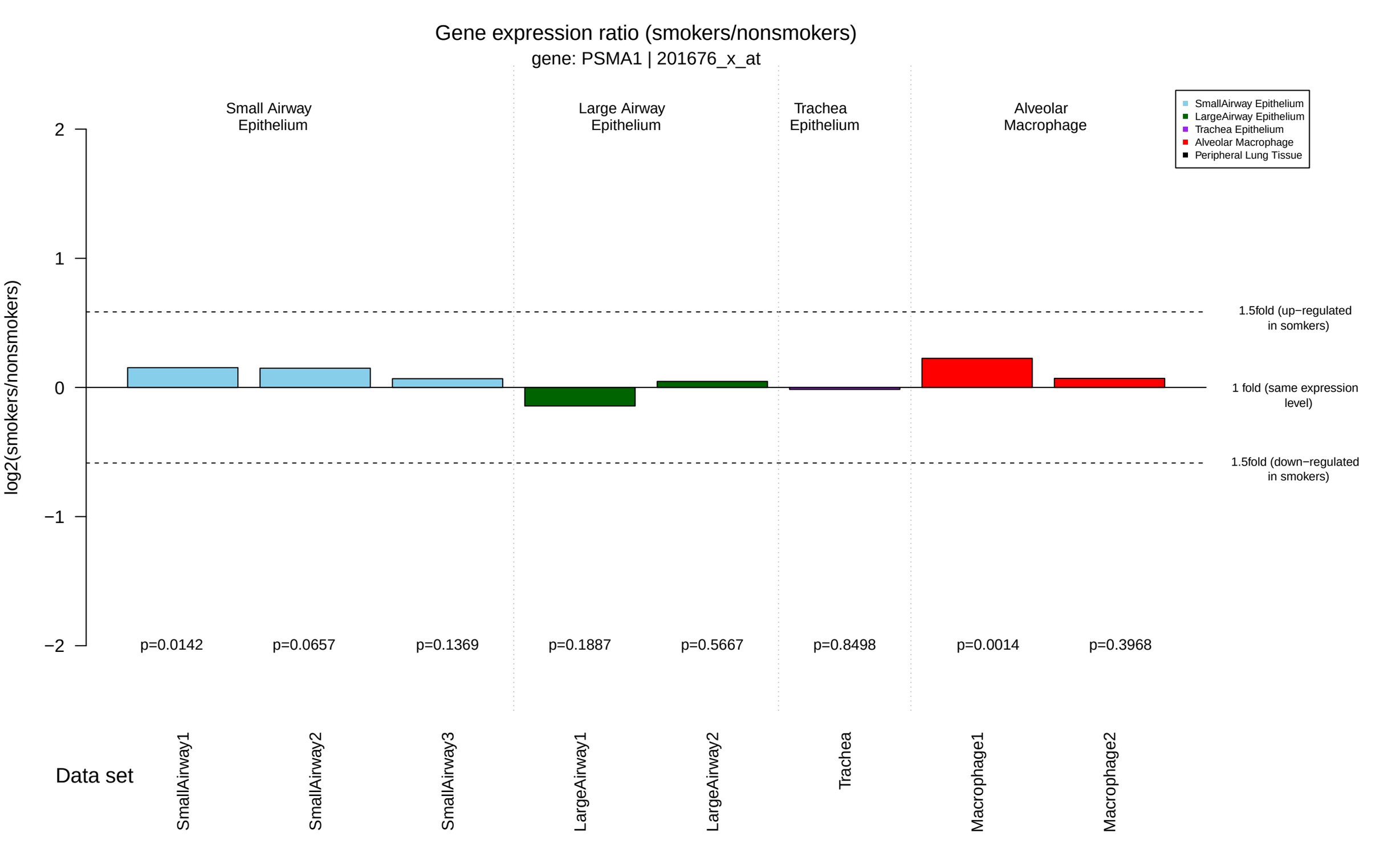
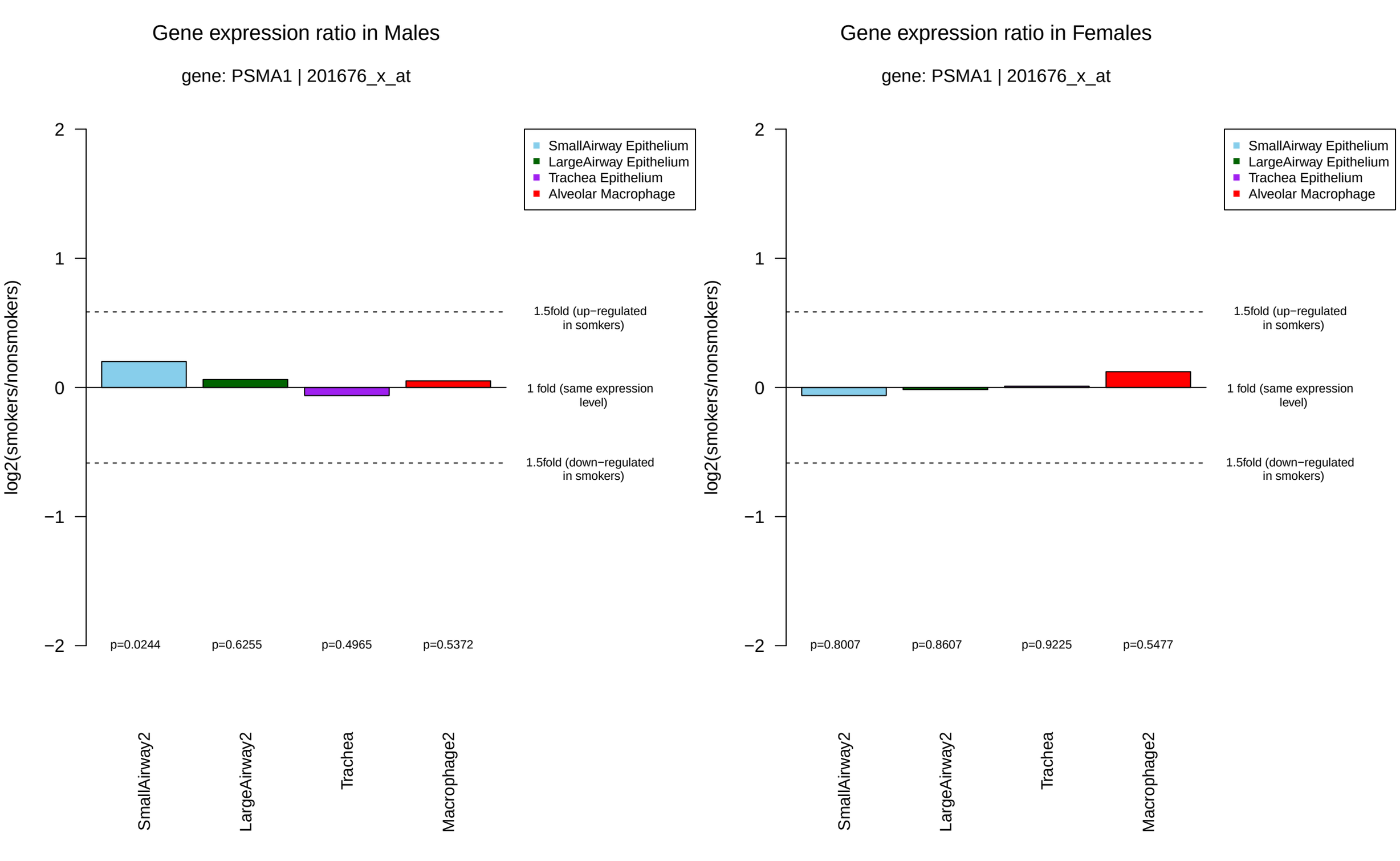
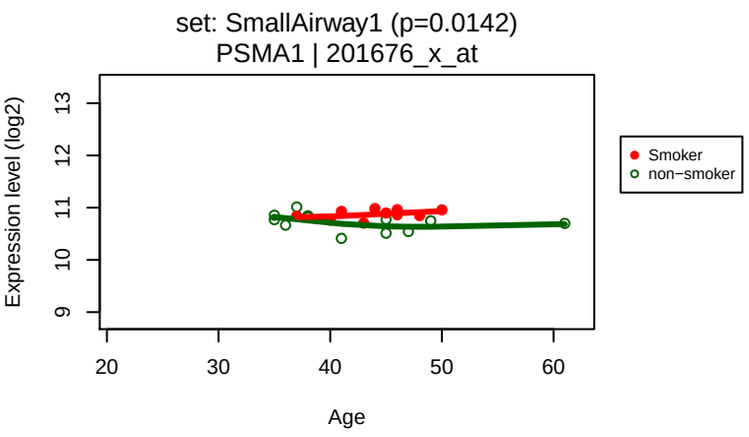
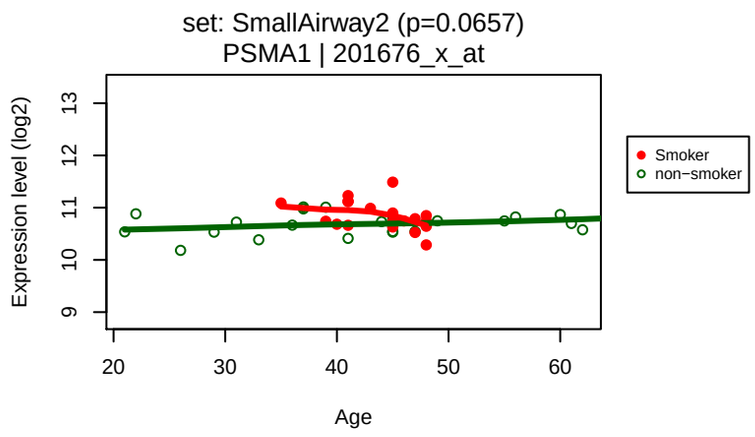
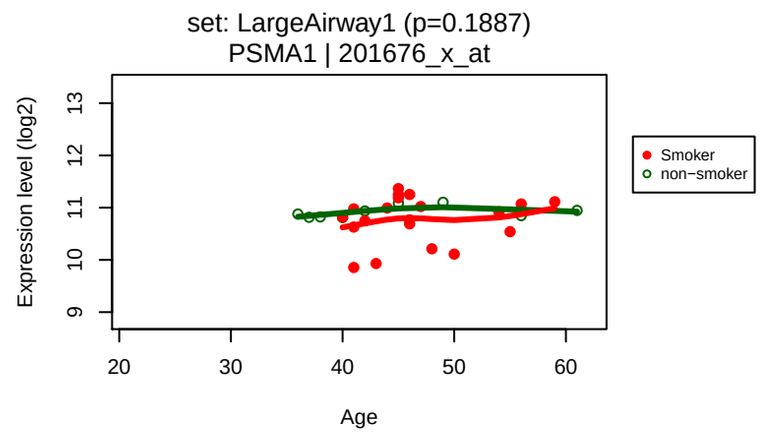

Gene expression of smokers and nonsmokers
gene: PSMA1 | 201676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0142
p=0.0657
p=0.1369
p=0.1887
p=0.5667
p=0.8498
p=0.0014
p=0.3968
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

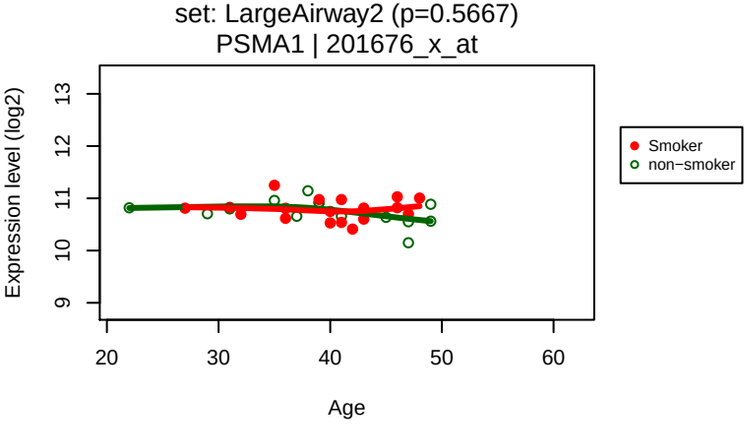

Gene expression ratio (smokers/nonsmokers)
gene: PSMA1 | 201676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0142
p=0.0657
p=0.1369
p=0.1887
p=0.5667
p=0.8498
p=0.0014
p=0.3968
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

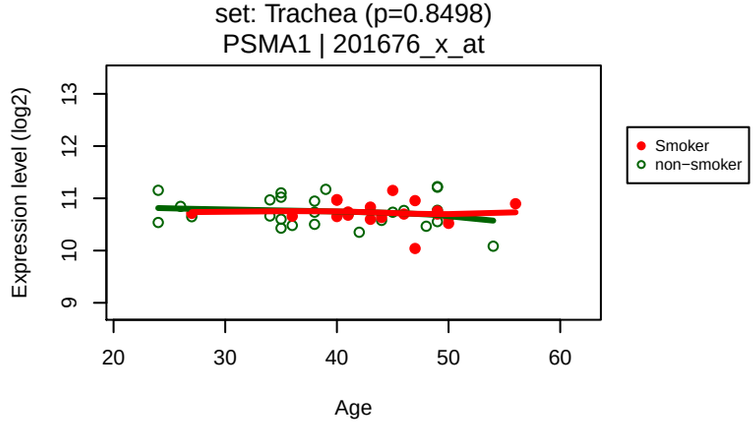

Gene expression ratio in Males
gene: PSMA1 | 201676_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0244
p=0.6255
p=0.4965
p=0.5372
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA1 | 201676_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8007
p=0.8607
p=0.9225
p=0.5477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

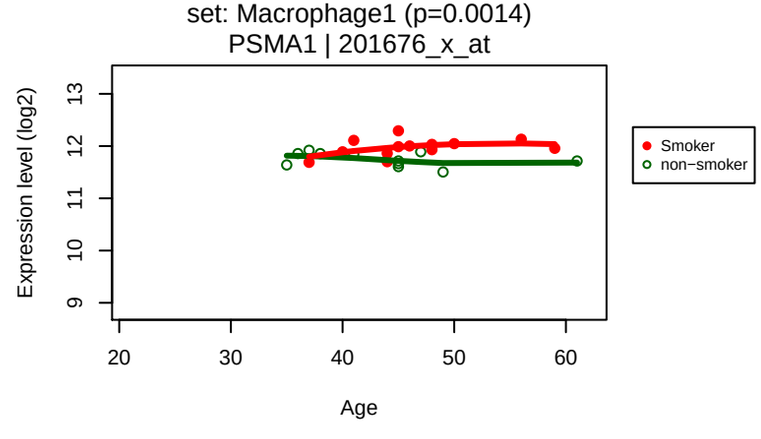

set: SmallAirway1 (p=0.0142)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker

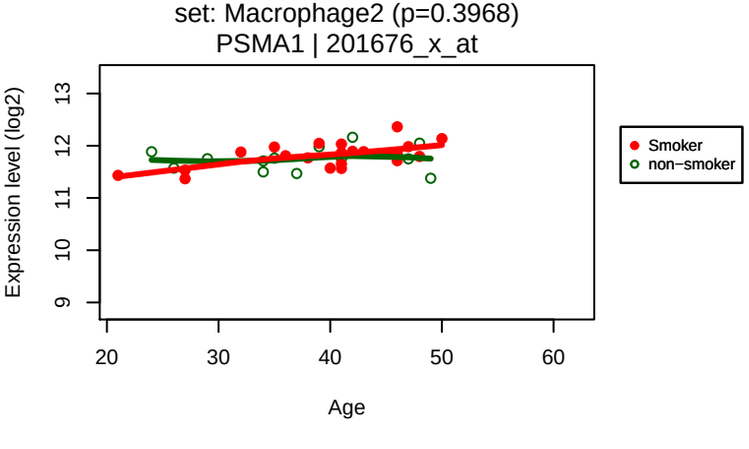

set: SmallAirway2 (p=0.0657)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker

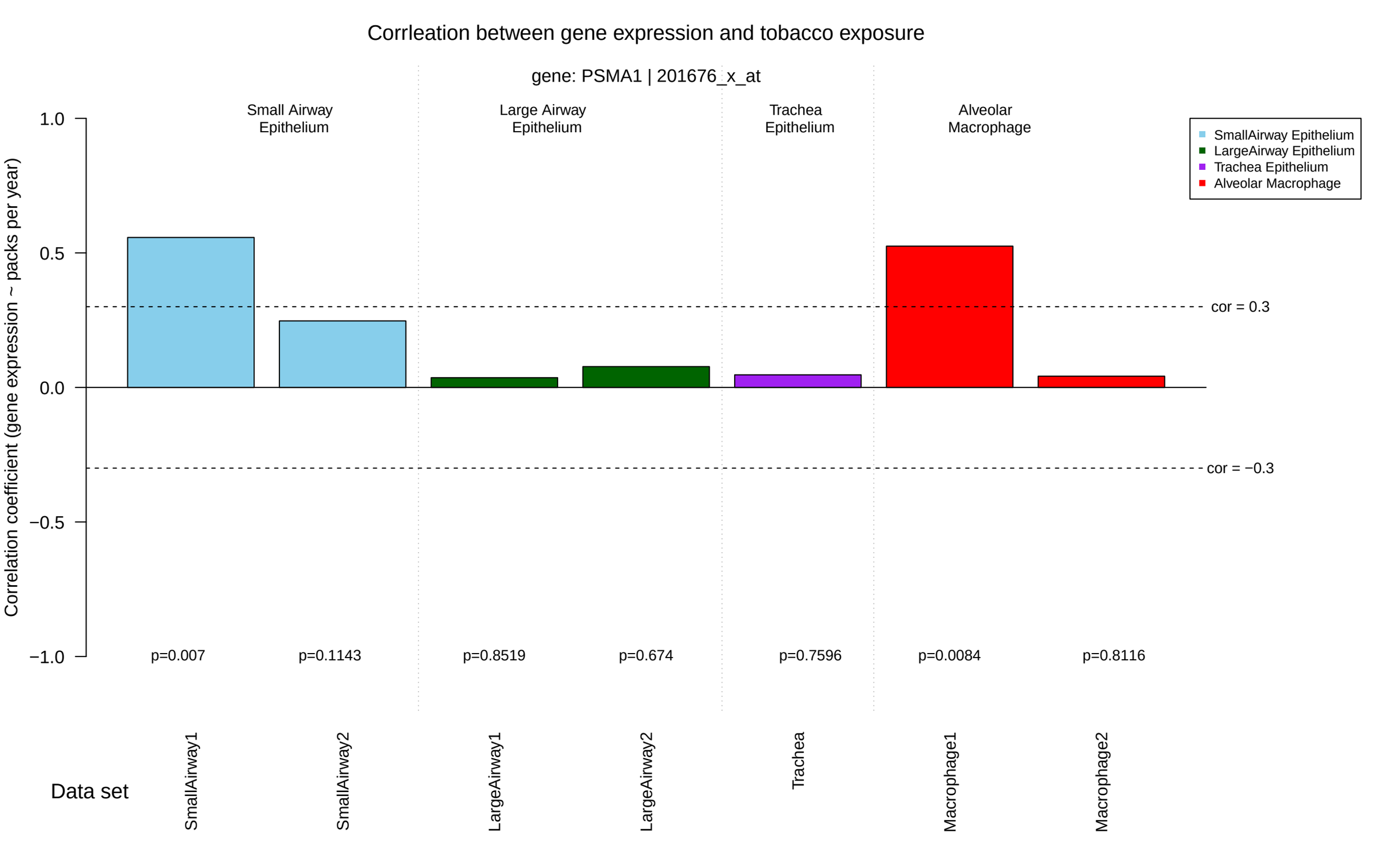

set: LargeAirway1 (p=0.1887)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5667)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8498)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0014)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3968)
PSMA1 | 201676_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA1 | 201676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.007
p=0.1143
p=0.8519
p=0.674
p=0.7596
p=0.0084
p=0.8116
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2